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Discount factors and monetary policy: Evidence from dual-listed stocks

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1 Big picture

1.1 Research question

The paper studies how *monetary policy* transmits to *equity prices* through investors' *discount factors* (risk-free discount rates and risk premia), rather than through changes in expected cash flows.

1.2 Core identification idea in one sentence

Use the fact that the *same firm* is traded simultaneously in Mainland China (A-shares) and Hong Kong (H-shares), so that the *A/H price ratio* nets out cash-flow news and isolates changes in *relative discount factors* across the two segmented investor bases.

1.3 Headline empirical finding

US monetary policy surprises move the A/H ratio strongly, but essentially only for *cycle-amplifying surprises*:

- **tightening-cycle surprises that are hikes**, and
- **easing-cycle surprises that are cuts**.

A 100bp cycle-amplifying surprise generates about a 0.30 change in the A/H ratio within a few days (Table 3–4; Fig. 4–5).

2 Institutional setting and why dual listing is useful

2.1 Same cash flows, different investor discount factors

A-shares and H-shares are (by construction in these dual listings) claims on the *same underlying firm cash flows* with equal voting and cash-flow rights. Therefore, absent frictions, they should have the same present value.

In reality, the two listings are traded in *segmented markets*:

- Mainland China is characterized by capital controls and a domestic investor base;
- Hong Kong has an internationally connected financial system and (crucially) a currency regime closely tied to the USD.

Thus, US monetary policy can pass through to Hong Kong discount rates more directly than to Mainland China discount rates.

2.2 Why the A/H ratio is a clean object

The usual challenge in estimating monetary policy effects on stock prices is that policy surprises may move prices through:

1. a **cash-flow channel** (aggregate demand or “information” about growth), and

2. a **discount-factor channel** (risk-free rates and/or risk premia).

The A/H ratio is designed to difference out (most) cash-flow news because the two shares are claims on the same cash flows.

3 Model: prices, discount factors, and the A/H ratio

3.1 Pricing equations (Eq. (1))

Let $J \in \{A, H\}$ index the trading region. Let m_A and m_H denote the monetary policy stances in Mainland China and Hong Kong, respectively. For firm i , the paper writes (following standard asset pricing notation):

$$P_A^i = \frac{1}{R_A^i(m_A)} \mathbb{E}[x^i(m_A, m_H)], \quad P_H^i = \frac{1}{R_H^i(m_H)} \mathbb{E}[x^i(m_A, m_H)]. \quad (1)$$

Interpretation.

- P_J^i is the price of firm i in region J .
- $R_J^i(m_J)$ is the *risk-adjusted discount factor* applied by investors in region J (it can be thought of as a gross discount rate in reduced form).
- $\mathbb{E}[x^i(m_A, m_H)]$ is the market expectation of firm i cash flows, which can depend on both policy stances in principle.

3.2 Discount factor decomposition

The paper decomposes the discount factor into a risk-free component and a risk premium component:

$$R_J^i(m_J) = r^f(m_J) + r_J^{p,i}(m_J), \quad J \in \{A, H\}. \quad (1)$$

Interpretation. Monetary policy can move the risk-free term r^f and/or the risk premium term r^p .

3.3 Key step: ratio cancels cash flows (Eq. (2))

Taking the ratio of Eq. (1) across the two markets:

$$\frac{P_A^i}{P_H^i} = \frac{\frac{1}{R_A^i(m_A)} \mathbb{E}[x^i(m_A, m_H)]}{\frac{1}{R_H^i(m_H)} \mathbb{E}[x^i(m_A, m_H)]} = \frac{R_H^i(m_H)}{R_A^i(m_A)}. \quad (2)$$

Thus the paper motivates:

$$\frac{P_A^i}{P_H^i} = \frac{R_H^i(m_H)}{R_A^i(m_A)}. \quad (2)$$

Economic meaning. Changes in the A/H ratio around US monetary policy announcements can be interpreted as changes in *relative discount factors* (Hong Kong vs Mainland China), not cash-flow revisions.

4 Empirical strategy: event study as difference-in-differences

4.1 Main regression (Eq. (3))

The baseline specification is:

$$\left(\frac{P_A}{P_H}\right)_{ist} = \alpha_i + \eta_s + \lambda_t + \beta \text{ Surprise}_s \times \text{Post}_t + \varepsilon_{ist}. \quad (3)$$

Indexing. i is a firm, s is an FOMC announcement, and t is an event-time day within the window around the announcement.

Fixed effects and what they do.

- α_i (company FE): absorbs time-invariant differences in the A/H premium across firms (e.g., persistent segmentation intensity).
- η_s (announcement FE): absorbs anything common to all firms on a given FOMC event date (macro announcements, global news, etc.).
- λ_t (event-time FE or post indicator): absorbs systematic patterns in event time (e.g., market drift in the days after an announcement).

Interpretation of β . β is the semi-elasticity in *levels*: it is the change in the A/H ratio induced by a 1 percentage point surprise in the target rate, comparing post-announcement to pre-announcement days.

4.2 Validating the monetary-policy shock in the relevant markets (Eq. (4))

The paper first checks that US monetary policy surprises actually move Hong Kong interest rates (but not Mainland China interbank rates). The event-study specification is:

$$\text{HIBOR}_{st} = \eta_s + \lambda_t + \sum_{\tau=-4}^5 \beta_{\tau}^{HB} \text{ Surprise}_s \times \mathbf{1}(\text{TimeSinceAnnouncement}_{st} = \tau) + \varepsilon_{st}. \quad (4)$$

Interpretation. The coefficients β_{τ}^{HB} trace the dynamic pass-through of an FOMC surprise into 3-month HIBOR (or HIBOR futures, or other rates).

4.3 Event-study dynamics for the A/H ratio (Eq. (5))

At the company-announcement level:

$$\left(\frac{P_A}{P_H}\right)_{ist} = \alpha_i + \eta_s + \lambda_t + \sum_{\tau=-4}^5 \beta_{\tau} \text{ Surprise}_s \times \mathbf{1}(\text{TimeSinceAnnouncement}_{ist} = \tau) + \varepsilon_{ist}. \quad (5)$$

What I learn from Eq. (5). This is both a pre-trend test (coefficients for $\tau < 0$) and a dynamics plot (how quickly the A/H ratio responds and whether it mean-reverts).

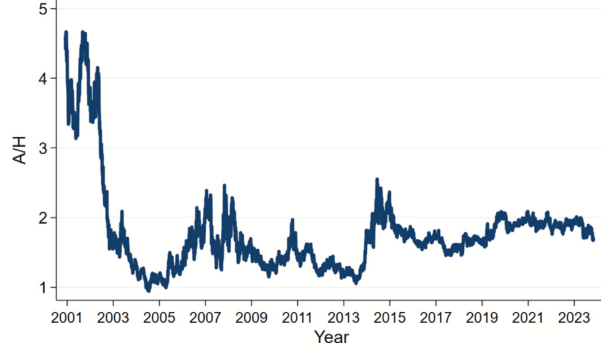


Fig. 1. Time series of median A/H ratio.

5 Data: what is measured

5.1 Dual-listed stocks and prices

The sample consists of dual-listed A/H firms traded in both markets, and the key outcome is the daily A/H price ratio.

5.2 Monetary policy surprises

The surprise $Surprise_s$ is the Kuttner (2001) surprise component of changes in the Fed target rate (computed from Fed funds futures around announcements). The baseline sample uses 129 FOMC announcements (Table 2), excluding zero-lower-bound episodes in the main specifications.

5.3 Event window

Around each announcement, the paper keeps a window of 5 consecutive trading days before and after (event time $t \in \{-4, \dots, 5\}$).

6 Main results: tables and figures, in order

6.1 Fig. 1 and Table 1: the A/H premium exists and is volatile

- **Fig. 1** shows the time series of the median A/H ratio (2001–2024). The ratio is far from 1 and varies substantially over time, motivating the need to understand what drives it.
- **Table 1 (Panel A)** reports that the A/H ratio has a mean around 2.34 with substantial dispersion across firm-announcement observations.
- **Table 1 (Panel B)** summarizes the monetary policy variables (target rate, surprises, HIBOR, HIBOR futures) on announcement dates.
- **Fig. 2** decomposes the sample into interest-rate hike, cut, flat, and zero-lower-bound regimes. This regime classification is crucial for the “cycle-amplifying” results later (Table 4).

Table 1
Summary statistics.

	Mean	SD	25th Pct.	Median	75th Pct.
Panel A: Company-announcement level					
A	15.99	40.25	3.884	7.058	14.55
ΔA	-0.0276	1.106	-0.0864	-0.00365	0.0663
H	8.354	16.47	1.892	3.608	8.096
ΔH	-0.000765	0.780	-0.0530	-0.00203	0.0437
A/H	2.339	1.882	1.415	1.818	2.663
$\Delta A/H$	-0.00637	0.116	-0.0278	-0.000424	0.0229
N	8496				
Panel B: Announcement level					
Target rate	2.878	1.862	1.250	2.250	4.875
Δ Target rate	0.00291	0.286	0	0	0.250
Surprise	-0.0151	0.0929	-0.00999	0	0.00484
3-Month HIBOR	2.683	1.701	1.170	2.261	4.209
Δ 3-Month HIBOR	-0.0154	0.103	-0.0301	0	0.0236
3-Month HIBOR future	2.732	1.684	1.250	2.250	4.250
Δ 3-Month HIBOR future	-0.00574	0.107	-0.0400	0	0.0300
N	129				

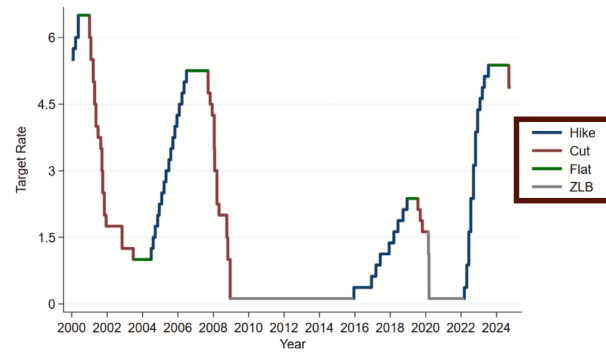


Fig. 2. Interest rate cycle.

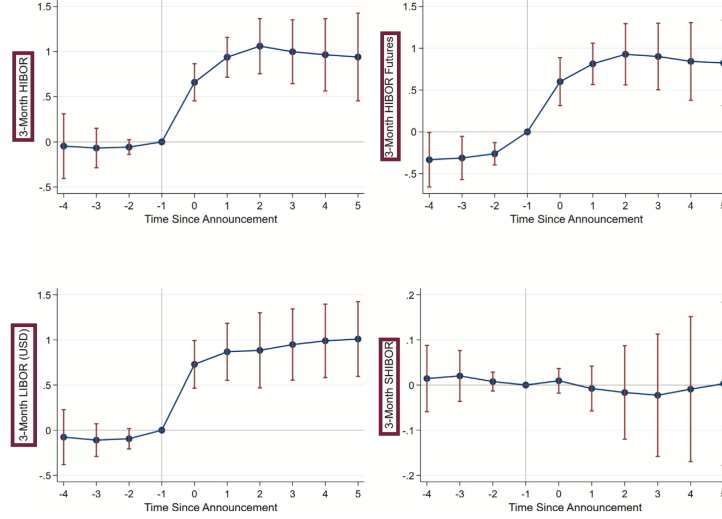


Fig. 3. Policy announcements, HIBOR, LIBOR, and SHIBOR.

6.2 Fig. 3: US policy surprises pass through to Hong Kong rates, not Mainland China rates

What **Fig. 3** does. It estimates Eq. (4) using different short rates:

- 3-month HIBOR and 3-month HIBOR futures (Hong Kong),
- USD LIBOR (global USD funding benchmark),
- SHIBOR (Mainland China).

Key interpretation.

- HIBOR and HIBOR futures respond strongly and quickly after FOMC announcements: evidence of pass-through of US monetary policy surprises to Hong Kong monetary conditions.
- SHIBOR shows little response: consistent with capital controls and independent Mainland monetary policy.

This supports the core assumption that discount rates in Hong Kong react to US policy shocks while Mainland discount rates do not (at least in the short run around announcements).

6.3 Table 2: the event set

Table 2 lists the FOMC announcements used (dates, target rates, Kuttner surprises). For me, this table is mostly a *data audit*:

- the paper has a long span (June 2000–Sept 2024),
- surprise magnitudes include both large crisis-era moves and smaller modern changes,
- the baseline sample size of announcements matches the regressions (typically 129).

Table 2

List of FOMC announcements.

Announcement date	Target rate	Kuttner surprise	Announcement date	Target rate	Kuttner surprise	Announcement date	Target rate	Kuttner surprise
2000/06/28	6.5	-.02	2005/08/09	3.5	0	2017/06/14	1.125	.009
2000/08/22	6.5	-.017	2005/09/20	3.75	.015	2017/07/26	1.125	0
2000/10/03	6.5	0	2005/11/01	4	0	2017/09/20	1.125	0
2000/11/15	6.5	0	2005/12/13	4.25	0	2017/11/01	1.125	0
2000/12/19	6.5	.052	2006/01/31	4.5	0	2017/12/13	1.375	.004
2001/01/03	6	-.382	2006/03/28	4.75	0	2018/01/31	1.375	-.005
2001/01/31	5.5	.005	2006/05/10	5	-.007	2018/03/21	1.625	.008
2001/03/20	5	.056	2006/06/29	5.25	-.015	2018/05/02	1.625	0
2001/04/18	4.5	-.425	2006/08/08	5.25	-.04	2018/06/13	1.875	-.004
2001/05/15	4	-.078	2006/09/20	5.25	0	2018/08/01	1.875	-.003
2001/06/27	3.75	.05	2006/10/25	5.25	0	2018/09/26	2.125	0
2001/08/21	3.5	.016	2006/12/12	5.25	0	2018/11/08	2.125	.003
2001/10/02	2.5	-.069	2007/01/31	5.25	0	2018/12/19	2.375	.032
2001/11/06	2	-.1	2007/03/21	5.25	0	2019/01/30	2.375	0
2001/12/11	1.75	0	2007/05/09	5.25	0	2019/03/20	2.375	.014
2002/01/30	1.75	.015	2007/06/28	5.25	0	2019/05/01	2.375	-.021
2002/03/19	1.75	-.026	2007/08/07	5.25	.026	2019/06/19	2.375	.041
2002/05/07	1.75	0	2007/08/17	5.25	.155	2019/07/31	2.125	.045
2002/06/26	1.75	0	2007/09/18	4.75	-.15	2019/09/18	1.875	-.033
2002/08/13	1.75	.034	2007/10/31	4.5	-.02	2019/10/30	1.625	-.017
2002/09/24	1.75	.025	2007/12/11	4.25	.008	2019/12/11	1.625	0
2002/11/06	1.25	-.194	2008/01/22	3.5	-.741	2020/01/29	1.625	.015
2002/12/10	1.25	0	2008/01/30	3	-.095	2022/03/16	.375	-.027
2003/01/29	1.25	.005	2008/03/18	2.25	.167	2022/05/04	.875	-.015
2003/03/18	1.25	.048	2008/04/30	2	-.055	2022/06/15	1.625	.036
2003/05/06	1.25	.037	2008/06/25	2	-.03	2022/07/27	2.375	-.031
2003/06/25	1	.15	2008/08/05	2	-.006	2022/09/21	3.125	-.043
2003/08/12	1	0	2008/09/16	2	.059	2022/11/02	3.875	-.002
2003/09/16	1	0	2008/10/08	1.5	-.142	2022/12/14	4.375	0
2003/10/28	1	0	2008/10/29	1	-.06	2023/02/01	4.625	-.008
2003/12/09	1	0	2008/12/16	.125	-.119	2023/03/22	4.875	.045
2004/01/28	1	0	2015/12/16	.375	.021	2023/05/03	5.125	.028
2004/03/16	1	0	2016/01/27	.375	0	2023/06/14	5.125	-.017
2004/05/04	1	-.006	2016/03/16	.375	-.01	2023/07/26	5.375	0
2004/06/30	1.25	-.01	2016/04/27	.375	0	2023/09/20	5.375	-.015
2004/08/10	1.5	.022	2016/06/15	.375	0	2023/11/01	5.375	0
2004/09/21	1.75	.017	2016/07/27	.375	0	2023/12/13	5.375	-.003
2004/11/10	2	0	2016/09/21	.375	-.042	2024/01/31	5.375	-.003
2004/12/14	2.25	0	2016/11/02	.375	-.005	2024/03/20	5.375	0
2005/02/02	2.5	0	2016/12/14	.625	0	2024/05/01	5.375	0
2005/03/22	2.75	0	2017/02/01	.625	-.003	2024/06/12	5.375	0
2005/05/03	3	0	2017/03/15	.875	.005	2024/07/31	5.375	.01
2005/06/30	3.25	0	2017/05/03	.875	0	2024/09/18	4.875	-.106

Table 3

Main results.

	(1)	(2)	(3)	(4)	(5)
Surprise $\times$ Post	0.306*** (0.0298)	0.292*** (0.0296)	0.293*** (0.0297)	0.292*** (0.0298)	0.276*** (0.0314)
Company FE	No	No	Yes	Yes	Yes
Event time	No	Post	Post	FE	FE
Company Spec. Trend	No	No	No	Common	Separated
Announcements	129	129	129	129	129
Observations	93,414	93,414	93,414	93,414	93,414

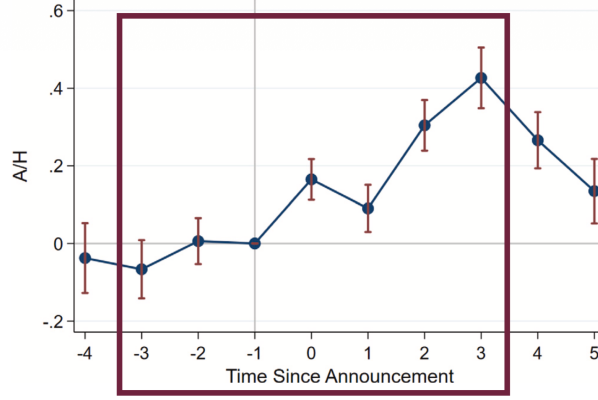


Fig. 4. Policy announcements and share prices.

6.4 Table 3 and Fig. 4: baseline A/H response

Table 3 (Eq. (3)). The coefficient on $Surprise \times Post$ is about 0.29 across specifications with different event-time controls and firm trends.

- **Sign.** A positive surprise (unexpected hike) increases the A/H ratio, consistent with H-shares falling relative to A-shares because Hong Kong discount rates rise.
- **Magnitude.** If $Surprise = -1$ (a 100bp surprise cut), the predicted effect is -0.29 , i.e., the ratio falls by about 0.29 because H-shares rise relative to A-shares.
- **Stability.** The coefficient is stable when adding company fixed effects and different event-time controls.

Fig. 4 (Eq. (5)). The event study shows:

- no evidence of pre-trends for $\tau < 0$ (important for identification),
- a clear post-announcement response that builds over a few days and peaks around day +3.

6.5 Table 4 and Fig. 5: asymmetry and “cycle-amplifying” surprises

Definition. A monetary policy surprise is *cycle-amplifying* if it *reinforces* the current policy direction:

- surprise **hike** during a **hiking** regime,
- surprise **cut** during a **cutting** regime.

A surprise is *cycle-contradicting* if it goes against the current regime.

Table 4. Splitting by both surprise direction and regime:

- In **hiking periods**, **surprise increases** are strongly significant (Panel A, col. 1).
- In **cutting periods**, **surprise decreases** are strongly significant (Panel B, col. 2).

Table 4

Main results split by surprise direction and interest rate cycle.

	(1) Increase	(2) Decrease	(3) Near zero
Panel A: Interest rate hike period			
Surprise $\times$ Post	0.390*** (0.112)	-0.0964 (0.107)	-0.459 (0.439)
Announcements	8	8	38
Observations	7535	9790	31,822
Panel B: Interest rate cut period			
Surprise $\times$ Post	0.0467 (0.179)	0.355*** (0.0400)	2.435 (3.726)
Announcements	13	19	9
Observations	5906	10,541	3882
Panel C: Interest rate flat period			
Surprise $\times$ Post	-0.351 (0.228)	-0.521 (0.528)	-0.630 (0.702)
Announcements	5	5	24
Observations	3575	3419	16,944

- Other combinations (including “contradictory” surprises and flat periods) are not statistically significant.

Interpretation. The discount-factor response in Hong Kong is not symmetric; it is concentrated in surprises that amplify the perceived monetary policy cycle.

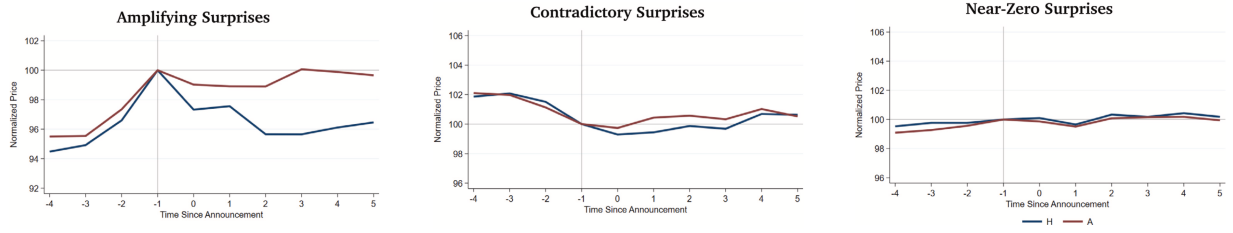


Fig. 5. Policy announcements and weighted normalized means of share prices split by surprise type.

Fig. 5. This figure shows raw normalized A- and H-share price paths around announcements, split into:

- amplifying surprises,
- contradictory surprises,
- near-zero surprises.

The central pattern: H-shares move strongly only for **amplifying surprises**, while A-shares move little in all cases. This is the “visual” version of Table 4.

6.6 Table 5: variance decomposition across announcements

What Table 5 does. It runs time-series regressions of the *average* A/H ratio (equal- or value-weighted) on surprise variables, to see whether the effect is primarily time-series (across announcements) rather than cross-sectional.

Interpretation.

Table 5

Variance decomposition: Time series variation.

	Equal-Weighted		Value-Weighted	
	Standardized (1)	Demeaned (2)	Standardized (3)	Demeaned (4)
Surprise $\times$ Post	0.185** (0.0772)	0.377** (0.157)	0.231*** (0.0534)	0.349*** (0.0806)
Surprise	-1.894 (1.442)	-3.852 (2.932)	-2.127 (1.485)	-3.211 (2.242)
Post	-0.00355 (0.00893)	-0.00721 (0.0182)	0.00216 (0.00772)	0.00326 (0.0117)
Adjusted R-squared	0.0256	0.0256	0.0324	0.0324
Announcements	129	129	129	129
Observations	1419	1419	1419	1419

- The coefficient on *Surprise* $\times$ *Post* remains positive and significant.
- This supports the idea that the effect is driven by announcement-time variation (monetary policy shocks) rather than changes in the composition of firms.
- The R^2 is low (around 0.03), consistent with many other forces driving the A/H ratio outside monetary policy surprises.

7 Cross-sectional heterogeneity: which stocks are most sensitive?

7.1 Triple-difference regression (Eq. (6))

To explore heterogeneity, the paper estimates:

$$\left(\frac{P_A}{P_H}\right)_{ist} = \alpha_i + \eta_s + \lambda_t + \beta_1 \text{Surprise}_s \times \text{Post}_t + \beta_2 \text{Charact}_{ist} + \beta_3 \text{Surprise}_s \times \text{Charact}_{ist} + \beta_4 \text{Post}_t \times \text{Charact}_{ist} + \beta_5 \text{Surprise}_s \times \text{Post}_t \times \text{Charact}_{ist} + \varepsilon_{ist}. \quad (6)$$

Interpretation of β_5 . It is the heterogeneity term: how much more (or less) sensitive the A/H ratio is to monetary policy surprises for stocks with higher values of the characteristic.

7.2 Table 6: interpretation panel-by-panel

Panel A: Price-to-earnings (PE) ratio (value vs growth). The estimated heterogeneity suggests stronger effects for *value* firms (low PE). In the split sample, below-median PE firms have roughly twice the response of above-median PE firms.

- A natural interpretation is that value firms have discount rates (or risk premia) that are more sensitive to macro/monetary shocks.

Panel B: Market capitalization (size). Smaller firms react more: the interaction term implies sensitivity declines with market cap. This is consistent with smaller firms being more exposed to discount-factor fluctuations (e.g., higher systematic risk or greater sensitivity to funding conditions).

Table 6

Main results interacted with characteristics.

	(1)	(2)	(3)	(4)	(5)	(6)
	Full sample	≤Median	>Median	Full sample	≤Median	>Median
Panel A: Price-to-Earnings			Panel B: Market capitalization			
Surprise × Post	0.301*** (0.0516)	0.346*** (0.0651)	0.162*** (0.0485)	2.518*** (0.529)	0.367*** (0.0575)	0.237*** (0.0346)
Charact	-0.0660 (0.0683)			-0.395*** (0.0474)		
Surprise × Charact	0.642 (0.395)			0.378*** (0.130)		
Post × Charact	-0.00825 (0.00749)			0.00238*** (0.000829)		
Surprise × Post × Charact	-0.161* (0.0908)			-0.0889*** (0.0210)		
Announcements	112	112	112	129	129	129
Observations	83,361	41,828	41,533	92,054	46,284	45,770
Panel C: CAPM β			Panel D: Lagged A/H Ratio			
Surprise × Post	-0.0368 (0.142)	0.220*** (0.0638)	0.381*** (0.0730)	0.160*** (0.0319)	0.211*** (0.0327)	0.396*** (0.0622)
Charact	0.518** (0.240)			0.0135 (0.0134)		
Surprise × Charact	-0.527 (1.373)			-0.514*** (0.0992)		
Post × Charact	-0.00337 (0.00888)			0.000156 (0.000169)		
Surprise × Post × Charact	0.362** (0.146)			0.0338*** (0.00650)		
Announcements	129	129	129	129	129	129
Observations	71,864	36,152	35,712	91,648	46,055	45,593

Table 7

Robustness.

	(1)	(2)	(3)	(4)
Surprise × Post	0.130*** (0.0254)	0.291*** (0.0288)	0.286*** (0.0279)	0.295*** (0.0300)
Set of announcements				
Include	ZLB Ann.			
Exclude		Zero Surprise Ann.	Holiday Ann.	Holiday Event Times
Announcements	201	58	116	116
Observations	160,885	40,766	85,741	78,030

Panel C: CAPM beta (systematic risk). Higher-beta stocks show larger responses, aligning with the idea that monetary policy surprises move not only the risk-free component but also *risk premia* that load more on high-beta assets.

Panel D: Lagged A/H ratio. Stocks with a higher lagged A/H ratio are more sensitive. *One interpretation is that the lagged premium is a proxy for persistent segmentation or discount-factor wedges, so shocks transmit more strongly where the wedge is already large.*

8 Robustness and alternative channels

8.1 Table 7: alternative event sets

Table 7 shows that the baseline effect is robust to:

- including ZLB announcements,
- excluding near-zero-surprise announcements,
- restricting to non-holiday announcements,
- redefining the event window around holidays.

Table A.1

Robustness interacting with US export share.

	Full sample					Share = 0	Share > 0
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Surprise $\times$ Post	0.213*** (0.0319)	0.199*** (0.0316)	0.200*** (0.0316)	0.201*** (0.0316)	0.196*** (0.0335)	0.194*** (0.0411)	0.208*** (0.0396)
US export share	-0.00644 (0.00835)	-0.00675 (0.00829)	0.0178 (0.0114)	0.0148** (0.00627)	0.0116** (0.00527)		
Surprise $\times$ US export share	-0.0662 (0.0466)	-0.0671 (0.0467)	-0.0199 (0.0339)	-0.0581** (0.0255)	-0.0225 (0.0211)		
Post $\times$ US export share	-0.0000270 (0.000254)	0.000537 (0.000372)	0.000531 (0.000370)	0.000607 (0.000380)	0.000606 (0.000388)		
Surprise $\times$ Post $\times$ US export share	-0.00155 (0.00601)	0.000126 (0.00607)	0.000174 (0.00608)	0.000319 (0.00609)	-0.000369 (0.00604)		
Company FE	No	No	Yes	Yes	Yes	No	No
Event time	No	Post	Post	FE	FE	Post	Post
Company Spec. Trend	No	No	No	Common	Separated	No	No
Announcements	110	110	110	110	110	110	92
Observations	53,988	53,988	53,988	53,988	53,988	29,135	24,853

The coefficient remains similar, supporting that the baseline result is not driven by mechanical calendar effects or a narrow subset of events.

8.2 Appendix A: are results driven by cash-flow exposure to the US? (Eq. (7) and Table A.1)

A potential concern is that some firms have more US revenue exposure, so US policy surprises could change expected cash flows differently across firms. The paper tests this using:

$$\left(\frac{P_A}{P_H}\right)_{ist} = \alpha_i + \eta_s + \lambda_t + \beta_1 \text{Surprise}_s \times \text{Post}_t + \beta_2 \text{USExportShare}_{ist} \quad (3)$$

$$+ \beta_3 \text{Surprise}_s \times \text{USExportShare}_{ist} + \beta_4 \text{Post}_t \times \text{USExportShare}_{ist} \quad (7)$$

$$+ \beta_5 \text{Surprise}_s \times \text{Post}_t \times \text{USExportShare}_{ist} + \varepsilon_{ist}.$$

Table A.1 interpretation. The triple interaction β_5 is not economically or statistically strong, and splitting the sample by whether US export share is zero yields similar estimates. This supports the idea that differential *cash-flow exposure* is unlikely to be the driver.

8.3 Appendix B: analysts' EPS forecasts (Fig. B.1–B.2 and Table B.1)

The paper further checks whether beliefs about future cash flows diverge across regions using analysts' EPS forecasts.

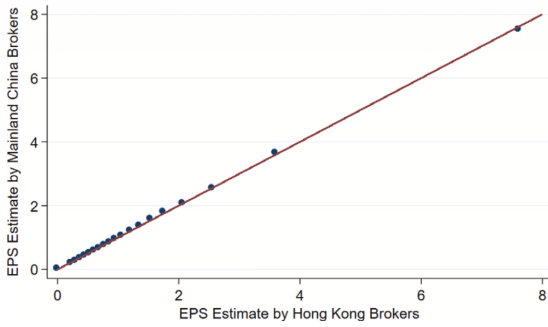


Fig. B.1. Binscatter plot of EPS forecasts.

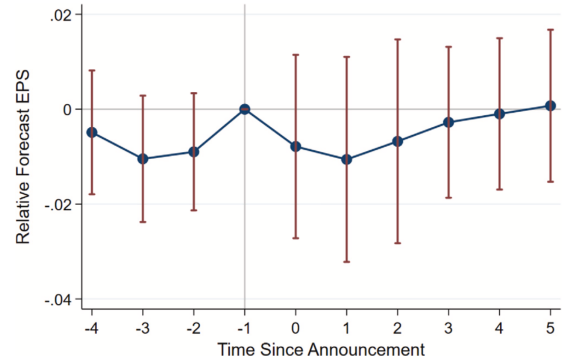
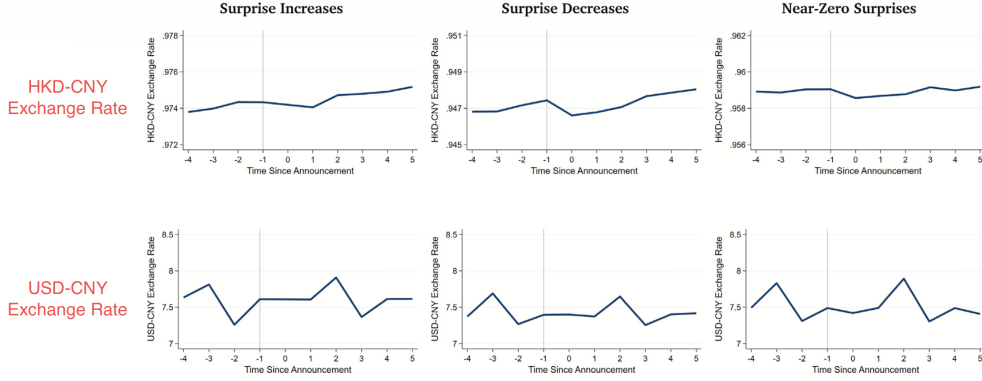


Fig. B.2. Policy announcements and EPS forecasts.

Table B.1

Monetary policy surprises and target rate and EPS forecasts.

	(1)	(2)	(3)	(4)	(5)
Surprise $\times$ Post	-0.00109 (0.00584)	0.00148 (0.00589)	0.00179 (0.00589)	0.00223 (0.00592)	0.00488 (0.00539)
Company FE	No	No	Yes	Yes	Yes
Event time	No	Post	Post	FE	FE
Company Spec. Trend	No	No	No	Common	Separated
Announcements	113	113	113	113	113
Observations	195,948	195,948	195,948	195,948	195,948

**Fig. C.1.** Means of exchange rates around surprise increases and decreases.

- **Fig. B.1** shows that Hong Kong and Mainland broker EPS forecasts are highly correlated (close to the 45-degree line).
- **Fig. B.2** shows no systematic change in relative EPS forecasts around FOMC surprises.
- **Table B.1** estimates an Eq. (3)-style regression with the relative EPS forecast ratio as the dependent variable and finds no significant effect of surprises.

Together, these reduce the plausibility of a “diverging cash-flow expectations” alternative.

8.4 Appendix C: exchange rates (Fig. C.1–C.3)

Because H-shares trade in HKD and A-shares trade in CNY, one might worry that exchange rate movements mechanically drive A/H ratios.

- **Fig. C.1** shows mean HKD–CNY and USD–CNY exchange rates around surprises (by direction bins).
- **Fig. C.2** provides event-study coefficients for spot exchange rates and finds limited immediate response.
- **Fig. C.3** repeats using 6-month forward rates and also finds no significant response.

This supports that exchange rate movements are not the main driver of the A/H response.

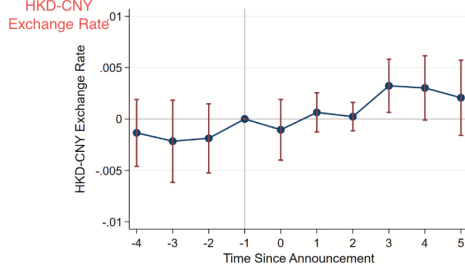


Fig. C.2. Policy announcements and exchange rates.

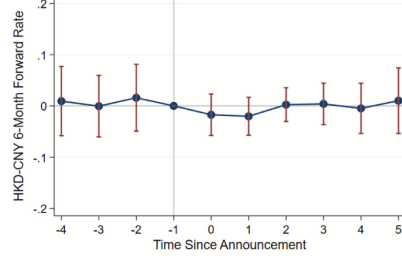


Fig. C.3. Policy Announcements and forward exchange rates.

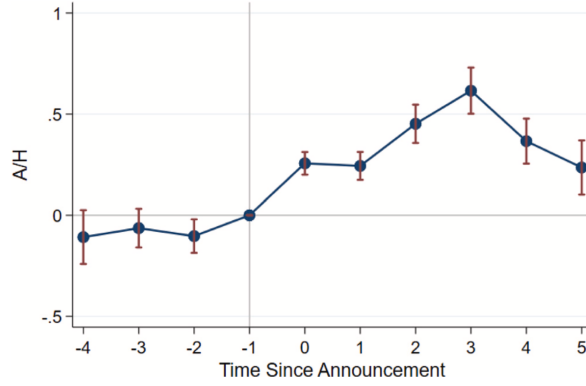
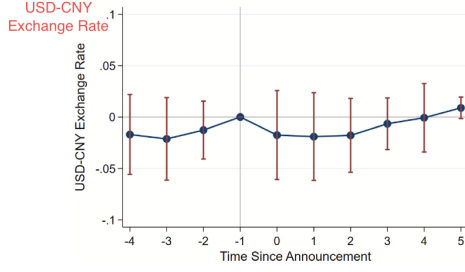


Fig. D.1. Stacked policy announcements and share prices.

8.5 Appendix D: dynamic and heterogeneous treatment effects (Eq. (8)–(9), Fig. D.1, Table D.1)

A modern DiD concern is bias from dynamic treatment effects or heterogeneous effects across units. The paper uses a *stacked* design (Baker et al.; Cengiz et al.):

Stacked main regression (Eq. (8)).

$$\left(\frac{P_A}{P_H} \right)_{i(T),s,t} = \alpha_{i(T),s} + \eta_{s,t} + \beta \text{Surprise}_{i(T),s} \times \text{Post}_t + \varepsilon_{i(T),s,t}. \quad (8)$$

Here $T = 1$ denotes the treatment group (the original event timing) and $T = 0$ is a control group constructed by lagging the same stock's price series.

Table D.1
Stacked main results.

	(1)	(2)	(3)	(4)
Surprise $\times$ Post	0.446*** (0.0405)	0.445*** (0.0404)	0.447*** (0.0408)	0.445*** (0.0409)
Company FE	No	Yes	Yes	Yes
Company Spec. Trend	No	No	Common	Separated
Announcements	129	129	129	129
Observations	186,232	186,232	186,232	186,232

Stacked event-study (Eq. (9)).

$$\left(\frac{P_A}{P_H}\right)_{i(T),s,t} = \alpha_{i(T),s} + \eta_{s,t} + \sum_{\tau=-4}^5 \beta_{\tau} \text{Surprise}_{i(T),s} \times \mathbf{1}(\text{TimeSinceAnnouncement}_{st} = \tau) + \varepsilon_{i(T),s,t}. \quad (9)$$

Table D.1 and Fig. D.1 interpretation. The stacked estimates remain positive and significant, and the event-study shape is similar. This suggests the baseline findings are not an artifact of problematic dynamic DiD behavior.¹

9 Summary

1. The A/H ratio is an empirical proxy for a *relative discount factor wedge* across segmented markets.
2. FOMC surprises shift Hong Kong discount rates (validated by HIBOR), but do not similarly shift Mainland rates (SHIBOR), which makes the A/H ratio respond.
3. The response is strongly *state-dependent*: it concentrates on policy surprises that *amplify* the prevailing rate cycle.
4. A battery of tests suggests the result is not primarily a cash-flow expectations story (US export shares, analyst EPS forecasts) nor an exchange-rate mechanical story.

10 Conclusion

10.1 Coherent summary: what the paper concludes

This paper isolates the *discount-factor channel* of monetary policy for stock prices by exploiting *dual-listed Chinese firms* whose A-shares trade in Mainland China and H-shares trade in Hong Kong. Because A- and H-shares are claims to the *same underlying cash flows* but trade in *segmented markets* (capital controls limit arbitrage across the two exchanges), changes in the A/H price ratio following policy news can be interpreted as changes in *investors' discount factors* rather than changes in expected cash flows.

The main conclusions can be stated in three connected points:

¹As noted in the paper, *Common* means controlling for firm-specific time trends that are common to both surprise increases and surprise decreases, whereas *Separated* means controlling for firm-specific time trends that are allowed to differ between surprise increases and surprise decreases.

1. **A large and economically meaningful discount-factor effect:** US monetary policy surprises move the relative price of H-shares versus A-shares, providing direct evidence that monetary policy affects stock prices through discount factors (beyond any cash-flow channel).
2. **Strong asymmetry by the policy cycle:** The effect is concentrated in *cycle-amplifying surprises*—surprise rate cuts during easing cycles and surprise rate hikes during tightening cycles—while cycle-contradicting surprises have much weaker or insignificant effects. The evidence is consistent with monetary surprises updating beliefs about the central bank’s reaction function and the associated pricing of risk.
3. **Persistence and measurement implications:** The impact is not confined to the highest-frequency window around announcements. The response persists after the initial announcement, implying that very high-frequency strategies or event studies that do not account for simultaneous information about cash flows may understate the magnitude of policy transmission to equity prices.

10.2 Economic intuition: why dual-listed shares identify the discount-factor channel

The identification hinges on three ingredients that link cleanly to asset-pricing logic.

Step 1: same cash flows, different discounting. A- and H-shares represent claims to the same firm-level cash flows, but they are priced by investors in two partially segmented markets. Under a common information environment, monetary policy news can shift *discount rates* differently across the two markets, even if expected cash flows are shared.

Step 2: differencing removes cash-flow news. Let the A/H ratio be:

$$\frac{P_{i,t}^A}{P_{i,t}^H}.$$

If cash-flow expectations move similarly in both markets, then the *ratio* differences out those revisions. What remains is a relative-discount-factor object: movements in the ratio reflect changes in how investors in the two markets discount the same cash flows.

Step 3: why US policy surprises matter in Hong Kong but not in Mainland China.

Because Hong Kong’s monetary/interest-rate environment is closely tied to the US via the exchange-rate regime, US monetary policy surprises plausibly shift discounting conditions for Hong Kong-traded assets. In contrast, Mainland China has an independent policy regime and capital controls, so the same US shock does not mechanically impose the same discount-rate adjustment on A-share investors. This asymmetry generates a difference-in-differences design: the same firm provides both “treated” and “control” claims to identical cash flows.

10.3 Why the effect is asymmetric (cycle-amplifying vs cycle-contradicting surprises)

A useful intuition is “belief updating about the central bank’s reaction function”:

- A **cycle-amplifying surprise** (e.g., an unexpectedly large hike during tightening) can be interpreted as a strong signal about policy resolve or responsiveness, inducing a sizable revision in risk premia and/or effective discount rates.
- A **cycle-contradicting surprise** (e.g., a cut during a tightening cycle) may be interpreted as noisy, transitory, or offset by contemporaneous information effects, leading to a weaker net revision in discount factors.

Thus the paper’s asymmetry result can be read as evidence that monetary announcements do not shift discount factors linearly; they shift them *state-dependently* through how investors interpret the policy signal.

10.4 Cross-sectional intuition: why riskier stocks react more

In standard asset-pricing logic, a change in discount factors (especially risk premia) has a disproportionately larger effect on assets that are more exposed to systematic risk. Consistent with this, the paper finds stronger responses for:

- high-beta stocks (greater market-risk exposure),
- small firms, and
- value firms (low valuation ratios),

which is naturally interpreted as a risk-premium (discount-factor) channel rather than a pure cash-flow channel.

10.5 Practical takeaway for researchers

This paper is a template for designing empirical strategies that isolate discount-rate variation:

1. Find two claims to (approximately) the same cash flows that are priced by different marginal investors.
2. Use a ratio or spread to difference out shared cash-flow news.
3. Use a policy shock that shifts discounting conditions in one investor base more than the other.

The result is an empirical “discount-factor wedge” that can be studied dynamically and in the cross section.

10.6 One-sentence takeaway

By comparing prices of the same firm’s stock traded in two segmented markets, the paper shows that monetary policy surprises move stock prices primarily through state-dependent revisions in investors’ discount factors, and the effect is large enough that conventional high-frequency approaches may understate policy transmission.